

Self-assessment questions on Forming by Plastic Deformation

- Indicate when hot forming is preferred in the industrial shaping of metals and alloys
- Explain which phenomena are exploited in hot forming and which are the relevant affecting factors
- Clear up the role played by the strain rate sensitivity coefficient
- Correlate the change of microstructure and mechanical properties during recrystallization of a TMCP
- Discuss why recrystallization is a convenient during hot rolling, especially if it is retarded to low temperature
- Prove why the final grain size after recrystallization decreases with increasing straining
- Elucidate when hot forging is favored over hot rolling in terms of product and mechanical properties requirements
- Compare the residual stress consequences resulting from small and large rolls during hot rolling of plates
- Compare hot vs. cold rolling in terms of thickness reduction, surface finish and final mechanical properties
- Underline the factors/performances controlled by normal and planar anisotropy in sheet forming
- Justify the characteristic values of normal anisotropy in Zn-alloys and Ti-alloys
- Based on the relevant sheet forming properties, prove why aluminum alloys are prone to forming failure
- Being aware that aluminium alloys are prone to failure in sheet forming, how this deficiency can be overcome
- Compare the forming ability of rimmed steels, Al-killed steels and interstitial free steels in terms of R

Self-assessment questions on Sheet forming

- Justify why the necking condition in sheet forming (2D) is delayed as compared to uniaxial plastic deformation
- Explain why lubrication affects only stretching but not drawing
- Clear up the strain condition (β) for which the sheet thickness and give a motivation
- Based on FLD, correlate the formability of Al-alloys vs. low carbon steels
- Elucidate how FLD curves are built from experiments
- Clarify how the sheet thickness affects the stretching and drawing formability
- Elucidate the factors affecting spring back of metals and alloys in tensile testing
- Illustrate which part of the FLD is affected by normal anisotropy
- Identify the formability parameters being determined by a sheet bending test
- Justify why the strain hardening coefficient affects both drawing and stretching in the FLD
- Based on FLD, explain both the least and the most critical strain condition (β) to ensure thickness uniformity
- Despite the simplicity of sheet bending test, which are the test outcomes of industrial interest
- Rank the following three metals in terms of bendability in the soft supply-condition and justify it: Al, Ti, Mg
- Explain why the electroforming process can affect the formability of Al-alloys and which FLD side is affected
- Specify the basic requirements for a given alloy to be formed superplastically